Rovonize Systems Research Sourcebook

Rovonize Systems Private Limited is a Kerala aerospace startup founded in 2018. The website project describes four areas of work: UAV development, engineering and precision manufacturing, aerospace research, and hands-on technical education. The older public website documents additional UAV variants and a history in precision agriculture. This sourcebook brings those records together for website writing and future company review. [S1–S5, W1, W3, E1]

What the sources cover

Company materials The supplied capability copy describes four engineering services and two training programmes. The supplied screenshot and earlier walkthrough identify the product lineup and Project AeroWeave. [S1–S3]

Website project Earlier Codex sessions and the React content files show how the material was organised and expanded for the redesign. Curriculum summaries, workflow copy and generated artwork in the build are editorial material. [S4–S5]

Older public website The live site and older indexed pages record products, published specifications, service lines, leadership profiles and contact details. Product availability and the applicability of specifications require company confirmation. [W1–W9]

Additional research Kerala Startup Mission and IIT Hyderabad's iTIC document the company's founding and incubation history. Public company announcements and contemporaneous agriculture reporting add milestones. [E1–E6]

Contents

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Company and leadership

Kerala Startup Mission records incorporation on 13 December 2018, identifies Rovonize as KSUM-recognised and lists aerospace and defence as its sector. Its profile records B2B and B2C models and lists drones, AI and machine learning, and manufacturing as technology areas. The older website describes integrated drone hardware, software and tailored applications for mapping, surveillance, agriculture, cargo delivery, defence and commercial or industrial operations. [E1, W1–W2]

The company presents its Make in India work in terms of domestic drone design and manufacturing, technical self-reliance, local talent, safety, quality and sustainability. Its older About page also describes bringing UAV technology to the wider community. [W1–W2]

Leadership named in company materials

Abison Varghese / Founder and Executive Chairman The company describes over 20 years of entrepreneurial experience, ventures including SecNeural and ThreatsView, CCIE Security certification and defence and government experience. He leads the company's work on AI-enabled autonomous aerial systems. [W1, E1, S3–S5]

Brig. P S Ramesh (Retd), PhD / Chief Executive Officer The company describes over two decades in UAS, doctoral work in UAS technology, patents and published research. It associates his work with military UAS design and the interaction of tactics, terrain and technology. [W1, S3–S5]

Arunkumar A R / Chief Technology Officer KSUM lists him as co-founder. His public bio cites more than 16 years in academia and UAVs, experience in aeronautical engineering and aircraft maintenance, and work in structural design and composites. He leads technology development. [W1, E1, S3–S5]

Dr. M Yogesh / Chief Strategy Officer His public bio describes a PhD, academic leadership, mechanical engineering and advanced systems research, with responsibility for strategy and innovation. The company also announced his appointment on LinkedIn. [W1, E5, S3–S5]

Historical team records include Gokul G.S., listed as a director in an iTIC profile that also listed Arunkumar as Managing Director, and Devan Chandrasekharan, identified in August 2020 reporting as a former CTO who went on to start Fuselage Innovations. [E2, E6]

Offices and enquiries

Kakkanad head office 3rd Floor, Confident Gem, Rajagiri Valley, Kakkanad, Ernakulam, Kerala 682039. KSUM adds unit 9/834-25 and records the location as the registered office. [W8, E1]

Research centre Rovonize Aerospace Research Centre, Memury, Pampakuda, Kerala 686667, as shown in the walkthrough and redesign. [S3, S5, E4]

General contact info@rovonize.com · +91 94970 86623 · +91 81295 25808. Both numbers appear on the older site's contact page. [W8, S3, S5]

UAV lineup and named variants

The redesign's four primary platforms come from the supplied screenshot and walkthrough. The public site's product records describe additional roles, variants and specifications. Each set of details below is tied to its own source. [S2–S5, W3–W6]

Spectra and STEM

Spectra Foldable quadrotor for reconnaissance. The draft site describes a foldable airframe for aerial observation and mission-specific integration. An approved specification sheet, operating envelope and release status are still needed. [S2–S5]

STEM Quadrotor for hands-on drone education and learning flight-system fundamentals. The walkthrough displays a Series 400 label and a flight-testing badge. Confirm the series designation, kit contents, availability and training use. [S2–S5]

Veeksha model records

The supplied material identifies Veeksha as a fixed-wing UAV for reconnaissance and surveillance. The older IBEX page also contains a separate block titled VEEKSHA RO V2.0, with a RO-V20 caption. It describes an intelligence, surveillance and reconnaissance (ISR) aircraft for defence, border security and law enforcement. Confirm whether this block describes the same aircraft shown in the redesign. [S2–S5, W5]

Older site specifications 6 kg maximum all-up weight, 1 kg payload, 5 km operating range and 2,000 m ceiling. The page lists day/night EO/IR imaging with HD zoom and night vision, a swappable Li-ion battery, autonomous and manual control, GPS waypoints, and encrypted RF, SATCOM and 4G/5G options. [W5]

Older site features and uses Long endurance, portability, low acoustic signature, AI-assisted object detection and tracking, and support for ISR sensors, LiDAR or custom payloads. It lists perimeter security, target tracking, search and rescue, battlefield monitoring and tactical support. [W5]

IBEX reconnaissance and logistics descriptions

The screenshot and walkthrough identify IBEX as a helicopter UAV for reconnaissance. The older website labels its IBEX logistics listing "Coming Soon" and describes a coaxial rotary-wing, internal-combustion platform intended for high-altitude cargo delivery. Confirm whether these are variants, different stages of one platform or different mission descriptions for the same aircraft. [S2–S5, W5]

Older site specifications Up to 100 kg modular payload and 250 kg maximum take-off weight (MTOW), 40 km range, 90 minutes endurance, 4,500 m launch altitude and 5,500 m ceiling, both above mean sea level (AMSL). [W5]

Older site features and uses The site claims a patented coaxial design for crosswinds, flight control for varying payloads, hybrid navigation in GPS-denied conditions, autonomous reverse-slope delivery and take-off, and encrypted control and video links. It lists military logistics, offshore cargo, disaster relief and urban delivery. Patent details and tested performance remain open. [W5]

Older products and agriculture

AgriHawk precision agriculture UAV

The walkthrough footer names Agrihawk. The old site has a dedicated AgriHawk page and lists it under Agriculture. It presents the aircraft for farmers, agronomists and researchers using high-resolution aerial imaging, multispectral and thermal sensing, AI-assisted analysis, GPS-guided flight, live data and interchangeable sensors or sprayers. It also advertises weather resistance, extended flight and fast charging. [S2–S3, W3–W4]

Published specifications 30 kg maximum all-up weight, 10 kg payload, 2 km operating range, 300 m ceiling, 4K or multispectral imaging, and a swappable 30,000 mAh battery. Flight modes include autonomous, manual and GPS waypoints. Links listed are 4G/5G, Wi-Fi and RF. [W4]

Uses listed Crop monitoring, precision spraying, irrigation management, yield estimation, weed and disease detection, and monitoring signs of pests, nutrient deficiency and water stress. The current model configuration and sales or support status are open. [W4]

EAGLE PRO

The old site describes a hybrid-power UAV with vertical take-off and landing (VTOL) and fixed-wing cruise. It lists autonomous waypoint flight, real-time telemetry, manual operation, encrypted RF, SATCOM and 4G/5G, quick assembly, and EO/IR, HD zoom, LiDAR or synthetic-aperture radar payload support. Applications include defence, border patrol, mapping, search and rescue, and cargo. [W6]

Published specifications 40 kg maximum all-up weight, 5 kg payload, 30 km operating range and 3,000 m ceiling. Its commercial status and relationship to the redesign lineup are open. [W6]

Other fixed wing VTOL records

Older Products page An unnamed 7 kg hybrid fixed-wing VTOL uses brushless DC motors for lift and thrust. Listed figures are 500 g payload, 3 km range, 1,000 m ceiling, 120 km/h maximum and 80 km/h cruise. The page prints endurance as “30 m”. It lists day/night cameras, return-to-home, parachute recovery, IRNSS navigation and GSM-assisted preprogrammed missions. [W3]

KSUM profile A separate H-configuration listing describes four brushless lift motors and a piston engine for thrust. It lists 30 kg all-up weight, 5 kg payload, over 200 km range and over six hours with the EFI engine, subject to fuel and payload, plus 4,000 m service ceiling, 120 km/h maximum speed and 70 km/h cruise. Model identity is open. [E1]

Palakkad field trial

The New Indian Express reported in August 2020 that a Rovonize-built agriculture drone sprayed micronutrients over three paddy fields covering 110 acres in Alathur. A monitoring agriculture officer reported a one-tonne-per-acre yield increase where the drone was used. The report does not establish an AgriHawk model identity or a repeatable yield outcome. [E6]

Engineering and manufacturing services

The company supplied this heading and introduction for the capabilities section. [S1]

Engineering & Technology Capabilities “From digital design to precision manufacturing, we transform engineering concepts into real-world solutions.”

Capabilities supplied by Rovonize

01 Composite manufacturing / Carbon fibre moulding Precision composite fabrication, mould development and lightweight structural components.

02 Electronics engineering / PCB design & manufacturing Custom PCB design, prototyping, fabrication and electronic assembly.

03 Precision manufacturing / CNC manufacturing Precision machining, custom components and rapid prototyping.

04 Digital engineering / 3D design & ANSYS simulation Advanced 3D modelling, product design, structural analysis and simulation.

05 Technical education / Engineering & technology courses Hands-on training in robotics, drones, electronics and technology. Includes 3- and 6-month internships in UAV Designing & Development and a 1-year Diploma in Drone Engineering & Artificial Intelligence. See page 7.

Supplied closing line “Design. Simulate. Manufacture. Innovate. Learn.” [S1]

UAV services on the older site

Custom drone design and development Application-specific UAVs for agriculture, defence, surveillance, logistics and industrial inspection. The site positions the service from concept through deployment and technical support. [W7]

Prototype development and testing Iterative design, rapid prototyping and testing of integrated hardware, software, sensors and communications. The services page discusses assessing stability, aerodynamics, battery life and payload capacity. [W7]

Aerial data and analytics support Capture, processing and analysis of aerial data, including GIS mapping, thermal imagery, multispectral analysis and AI-assisted processing. Example applications are agriculture, infrastructure monitoring, environmental studies and security. The old Products page also mentions IoT and AI image-processing services. [W3, W7]

Draft engineering workflow and technical leads

The redesign uses four stages: define and design, analyse and simulate, prototype and manufacture, then integrate and validate. The walkthrough also names CAD surfacing, aerofoil lofting, parametric ribbing, digital packaging, PCB surface-mount assembly, multilayer RF routing and redundant flight-control microcontrollers. The footer adds aerodynamic modelling, composite tooling, embedded flight-control firmware and hardware-in-the-loop simulation. Verify which activities the company offers directly and which belong to individual projects. [S3–S5]

Aerospace research and milestones

Project AeroWeave

AeroWeave is the named research project in the supplied material. Its stated focus is out-of-autoclave carbon prepreg and fabrication methods for lightweight structures. Prepreg is fibre reinforcement pre-impregnated with resin. The draft R&D page connects materials selection and structural behaviour with modelling, simulation, mould development and fabrication. [S2–S5]

Walkthrough claim Three identical cards describe a low-void vacuum consolidation curing schedule for high-modulus pitch-based carbon fibre and a claimed 22% weight reduction with equivalent torsional aero-damping. The baseline, specimen, test method and report are pending. The repeated cards represent one named project in the walkthrough. [S3, 14 seconds]

Startup history and public milestones

2018 / Formation KSUM records incorporation in December 2018. The iTIC directory describes Rovonize as a product business building and selling surveillance aircraft and helicopters for civil and military applications. [E1–E2]

2020 / Agriculture field work The Palakkad paddy trial documents an early agricultural application of a Rovonize-built drone. See page 4. [E6]

2021 / Surveillance UAV support IIT Hyderabad announced Rovonize among early-stage autonomous-navigation startups eligible for iTIC investment of up to ₹10 lakh each and identified its work on surveillance UAVs. The announcement gives an upper limit. Actual disbursement remains unverified. [E3]

2022–23 / Incubation records The iTIC annual report lists Rovonize under TiHAN EiR/PRAYAS and separately describes an ABCD programme concept to detect and intercept small drones in mid-air while avoiding damage on the ground. Development status and any link to the current portfolio remain open. [E2]

Publicly announced / Academic collaboration Rovonize announced a memorandum of understanding between Rovonize Systems Pvt. Ltd. through its Aerospace Research Centre and Noorul Islam Centre for Higher Education (NICHE). See page 7 for the stated scope. [E4]

Publicly announced / Quality management Rovonize announced ISO 9001:2015 certification on its company LinkedIn page. The walkthrough also displays an ISO badge and a MIL-STD-810H compliance badge. Certificate scope and validity, and product-specific military-standard test evidence, remain open. [E5, S3]

Research evidence to collect

For AeroWeave, record the project owner, stage, material system, process parameters, test methods, baseline comparisons, approved images and any demonstrable outcomes. For historical work, collect attributable programme references, flight-test or field-trial records and public case studies. [S2–S3, E2–E3]

Technical education and partnerships

Programmes supplied by the company

Rovonize Engineering Academy is the name used in the walkthrough and redesign for hands-on learning in robotics, drones, electronics and engineering technology. The company's supplied programme titles and durations are recorded below. [S1–S5]

3 months or 6 months / UAV Designing & Development Internship course in UAV design and development. The draft uses "UAV Design & Development" and proposes UAV fundamentals and aerodynamics, CAD and structural simulation, propulsion and electronics, fabrication, assembly and flight operations. [S1–S2, S5]

1 year / Diploma in Drone Engineering & Artificial Intelligence The draft proposes UAV design and analysis, avionics, sensors and communications, AI and autonomous navigation, manufacturing, prototyping and project work. Confirm the detailed syllabus, awarding body and recognition. [S1–S2, S5]

Earlier walkthrough curriculum

Six-month UAV Design & Development Course The video lists UAV fundamentals, aerodynamics and stability, CAD using Fusion 360 or SolidWorks, propulsion and electronics, fabrication and assembly, flight control and programming, flight operations and testing, and MATLAB or ANSYS simulation. [S3, 16–18 seconds]

Twelve-month Advanced UAV Systems & Applications Course The video lists flight mechanics, autonomous navigation and mission planning, Kalman-filter sensor fusion, test-range launch procedures, advanced UAV design, structural simulation, propulsion and power management, avionics, payload integration, VTOL and rotary-wing systems, fabrication and prototyping, flight trials and data analysis, case studies and an industry project. Its relationship to the supplied diploma is unconfirmed. [S3, 16–18 seconds]

Both video cards show an aerospace or mechatronics degree prerequisite and a September 2025 cohort. The video also mentions composite lab and wind tunnel access and claims more than 85% of fellows progress into full-time Rovonize development programmes. Treat these as draft-era statements pending current admissions information and evidence for the outcome claim. [S3]

Industry and academic collaboration

The NICHE MoU announcement names internships, student and capstone projects, R&D, product development, faculty development, technical workshops, seminars, certification courses, career guidance and placement support. It identifies UAS, robotics, embedded systems, AI, IoT, electronics and PCB design as collaboration areas. The announcement describes intended activities. [E4]

Admissions information still to collect

Collect the current syllabus and intake calendar, entry requirements by programme, delivery site and format, teaching hours, fees, application steps, instructor profiles, equipment and flight access, awarding body, assessment and credential details. The academy draft currently directs applicants to enquire for details. [S5]

Details to confirm for the website

Product identity and availability

The four redesign platforms can each be described at the airframe-and-use level. Request an approved sheet for each with model or version, development or availability status, configuration, mission, payload, range, endurance, altitude, test conditions and approved media. Resolve IBEX reconnaissance versus logistics, Veeksha versus RO V2.0, and the identity and status of AgriHawk, EAGLE PRO and the two older VTOL listings. [S2–S5, W3–W6, E1]

The walkthrough repeats 4.80 m wingspan, 85 kg MTOW, 18.5 hours endurance, 22 kg payload, MIL-SPEC T-41 and fixed-wing monocoque copy across all four cards, including quadrotor and helicopter labels. It marks Spectra and Veeksha “Production” and IBEX and STEM “Flight Testing”. These are draft records pending product-specific evidence. [S3, 10–14 seconds]

People and photographs

The walkthrough repeats one European Aerodynamics Institute biography, transonic-wing and stability-augmentation specialisms, an aeronautical engineering PhD and 18 years of experience for all four people. Use the individual company bios on page 2 as the public reference. Confirm titles, headshots and additional team members. Prior design sessions explicitly requested AI product placeholders and created prompts for engineering and lab illustrations. Use approved aircraft, facility and process imagery for claims about actual hardware or equipment. [S3–S5, W1]

Service scope and substantiation

Confirm in-house versus partner delivery, process and machine limits, materials, tolerances, quality controls, lead times and customer examples. For the ISO certificate, request the issuer, scope, dates and number. For military-standard or patent wording, request the applicable test report or patent reference and the configuration it covers. [S1, S3, W5, E5]

Careers and public channels

The older careers page lists on-site, full-time Electronics Engineer, HR & Administration Executive, Mobility Solutions, Manufacturing, Marketing and Production Engineer entries. Its general application channel is info@rovonize.com for several entries. A company LinkedIn post separately advertises a Kochi Social Media Executive role via jees@rovonize.com. Verify active vacancies and routing. The older site also links to its YouTube videos and carries a Blogs landing page. [W9, E5]

Website content from the design sessions

The project organises the main content around UAV Products, Engineering Services, R&D Lab, Academy and About, with Careers and Contact as distinct destinations or enquiry paths. Prior research used Freefly, Wingtra and Auterion for UAV presentation, Protolabs for manufacturing service structure, and Festo Didactic and Arduino Education for training presentation. These were design references for the redesign. [S4–S5]

Source register

Reference codes point to the supplied material, company publications or independent records. The public pages were reviewed on 9 November 2026. Older or indexed content can differ from current navigation.

Supplied material and project sessions

S1 Engineering & Technology Capabilities Company copy supplied in this request.

S2 Website section screenshot Attachment dated 9 November 2026.

S3 WhatsApp Video 2026-11-07 at 3.39.39 PM.mp4 22-second design walkthrough supplied in the earlier project session.

S4 Codex project sessions Create React app with Tailwind, Find international UAV website ideas, Update header navigation, Optimize image loading strategy, and Add app color variables.

S5 Rovonize website workspace Content files under src/data and the related company, engineering, research, academy and product pages.

Rovonize public website

W1 Homepage and leadership

W2 About Us

W3 Products and generic hybrid VTOL

W4 AgriHawk

W5 IBEX and embedded VEEKSHA RO V2.0 entry

W6 EAGLE PRO

W7 Services

W8 Contact Us

W9 Careers · Videos · YouTube · Blogs

Institutional records and announcements

E1 Kerala Startup Mission company profile

E2 iTIC annual report 2022–23 · Startup profiles and founders

E3 IIT Hyderabad announcement on autonomous navigation startups

E4 Rovonize announcement of the NICHE MoU

E5 Rovonize company profile and announcements on LinkedIn

E6 The New Indian Express report on the Palakkad trial